```
2000 'SUBROUTINE TO CALCULATE THE NEW RECURSION COEFFICIENTS
2010 'Variables entering routine:      A[ ], B[ ], DELTA, MU
2020 'Variables exiting routine:       A[ ], B[ ], EOLD, ENEW
2030 '
2040 GOSUB 3000                                 'FIND THE CURRENT ERROR
2050 EOLD = ER                                  'store current error in variable, EOLD
2060 '
2070                                            'FIND THE ERROR SLOPES
2080 FOR P% = 0 TO NP%                          'loop through each "a" coefficient
2090   A[P%] = A[P%] + DELTA                    'add a small increment to the coefficient
2100   GOSUB 3000                               'find the error with the change
2110   SA[P%] = (ER-EOLD)/DELTA                 'calculate the error slope, store in SA[ ]
2120   A[P%] = A[P%] - DELTA                    'return coefficient to original value
2130 NEXT P%
2140 '
2150 FOR P% = 1 TO NP%                          'repeat process for each "b" coefficient
2160   B[P%] = B[P%] + DELTA
2170   GOSUB 3000
2180   SB[P%] = (ER-EOLD)/DELTA                 'calculate the error slope, store in SB[ ]
2190   B[P%] = B[P%] - DELTA
2200 NEXT P%
2210 '                                          'CALCULATE NEW COEFFICIENTS
2220 FOR P% = 0 TO NP%                          'loop through each coefficient
2230   A[P%] = A[P%] - SA[P%] * MU              'adjust coefficients to move "downhill"
2240   B[P%] = B[P%] - SB[P%] * MU
2250 NEXT P%
2260 '
2270 GOSUB 3000                                 'FIND THE NEW ERROR
2280 ENEW = ER                                  'store new error in variable, ENEW
2290 '
2300 RETURN


3000 'SUBROUTINE TO CALCULATE THE FREQUENCY DOMAIN ERROR
3010 'Variables entering routine:      A[ ], B[ ], T[ ]
3020 'Variables exiting routine:       ER
3030 '
3040 FOR I% = 0 TO N%-1                         'LOAD SHIFTED  IMPULSE INTO IMX[ ]
3050  REX[I%] = 0
3060  IMX[I%] = 0
3070 NEXT I%
3080 IMX[12] = 1
3090 '                                          'CALCULATE IMPULSE RESPONSE
3100 FOR I% = 12 TO N%-1
3110  FOR J% = 0 TO NP%
3120    REX[I%] = REX[I%]  +  A[J%] * IMX[I%-J%]  +  B[J%] * REX[I%-J%]
3130  NEXT J%
3140 NEXT I%
3150 IMX[12] = 0
3160 '                                          'CALCULATE THE FFT
3170 GOSUB 1000                                 'Table 12-4, uses REX[ ], IMX[ ], N%
3180 '
3190                                            'FIND FREQUENCY DOMAIN ERROR
3200 ER = 0                                     'zero ER, to use as an accumulator
3210 FOR I% = 0 TO N%/2                         'loop through each positive frequency
3220   MAG = SQR(REX[I%]^2 + IMX[I%]^2)         'rectangular --> polar conversion
3230   ER = ER + ( MAG - T[I%] )^2              'calculate and accumulate squared error
3240 NEXT I%
3250 ER = SQR( ER/(N%/2+1) )                    'finish calculation of error, ER
3260 '
3270 RETURN
```

TABLE 26-5